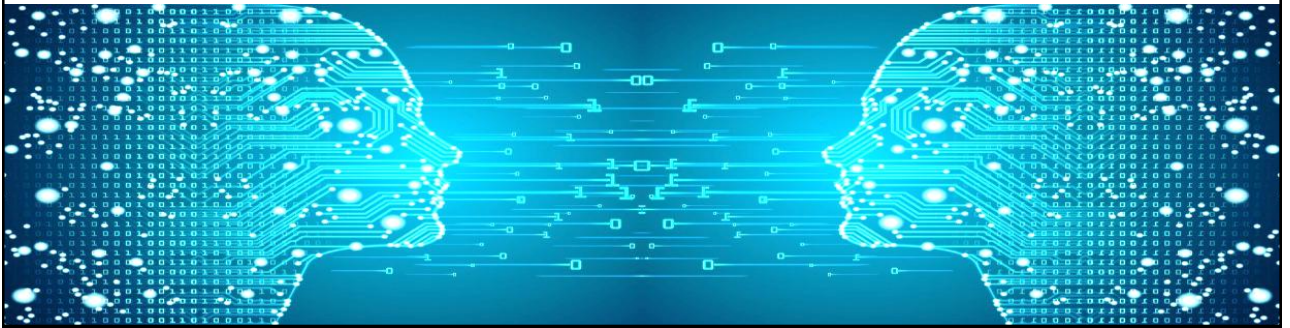
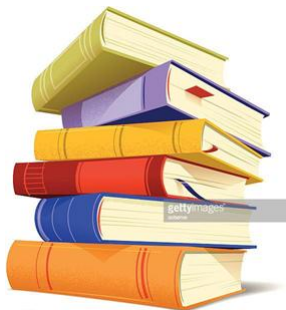


Machine Learning



Support Vector Machine (SVM)



- Introduction
- Problem Definition
- Design of the SVM
- Nonlinear Kernels

Introduction to SVM

- SVMs are supervised learning models with associated algorithms that:
 - Analyze data and recognize patterns.
 - Are used for **classification** and **regression**.
 - Build a model that assigns new examples into one of two categories.

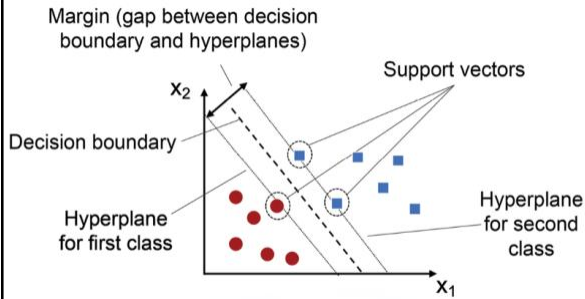
Problem Definition

- Given N vectors $\{x_1, x_2, \dots, x_n\}$, where each x_j has m features $\{a_{j1}, a_{j2}, \dots, a_{jm}\}$ and belongs to class C_1 or C_2 .
- The data is **linearly separable** if a hyperplane exists that separates the two classes:

$$w^T x + b = 0$$

- **SVM Objective:** Find the hyperplane that maximizes the margin between classes C_1 and C_2 .

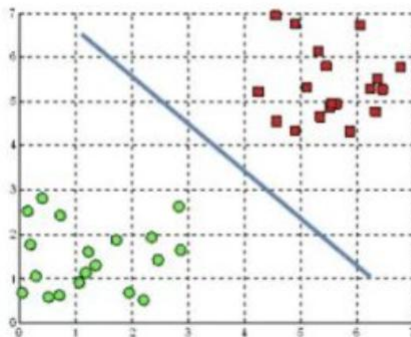
Support Vectors and Margin



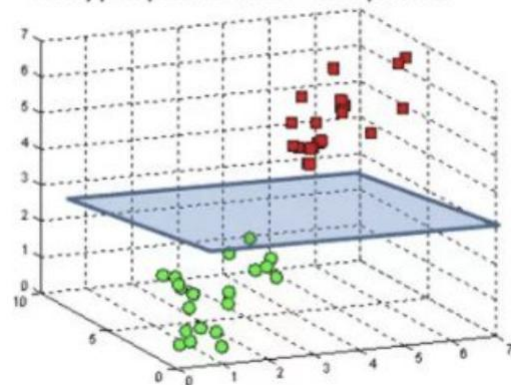
- Support vectors are the **closest data points** to the separating hyperplane.
- The **margin** is the distance between these points and the hyperplane.
- The optimal hyperplane maximizes this margin.
- Ref. Fig.8.1m Fig.8.2

Hyperplanes

A hyperplane in $\mathbb{R}^2$ is a line



A hyperplane in $\mathbb{R}^3$ is a plane



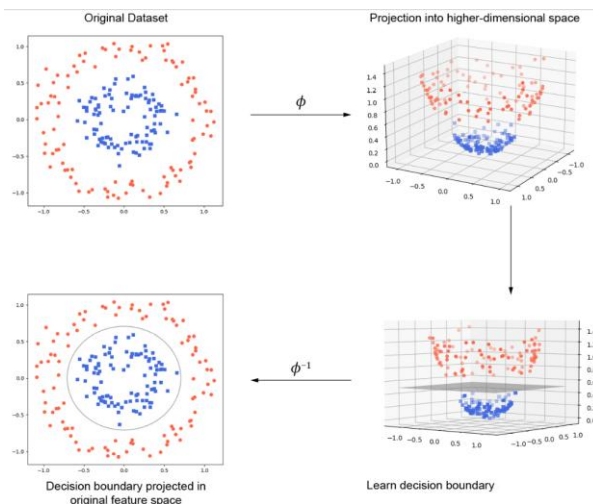
SVM Design

- Training involves adjusting weights (w) and bias (b) so that:

$$w^T x_i + b \geq 0 \text{ for class } C_1$$

$$w^T x_i + b < 0 \text{ for class } C_2$$

Nonlinear Kernel Case



- Some datasets are **not linearly separable**.
- SVM handles this using the **kernel trick**:

$\phi(x)$: input $\rightarrow$ higher-dimensional space

Common kernels

- Common kernels:
 - Linear: $K(x, x') = x^T x'$
 - Polynomial: $K(x, x') = (x^T x' + 1)^d$
 - RBF (Gaussian): $K(x, x') = \exp(-\gamma |x - x'|^2)$

Multi-class Classification using SVM

- Standard SVM is inherently a binary classifier. We need strategies to extend it to multi-class problems.
- Strategies for Multi-class SVM:
 1. One-vs-Rest (OvR)
 - Train one SVM per class.
 - Each SVM separates one class from the rest.
 2. One-vs-One (OvO)
 - Train an SVM for every pair of classes.
 - Total classifiers = $C(C-1)/2$ for C classes.

Decision is made based on which classifier outputs the highest confidence or wins the majority vote

One-vs-Rest Example

- Suppose we have 3 classes: A, B, C.
- Train 3 SVMs:
 - SVM1: A vs (B + C)
 - SVM2: B vs (A + C)
 - SVM3: C vs (A + B)
- For a new sample, select the class with the highest decision function value.

One-vs-One Example

- Suppose we have 3 classes: A, B, C.
 - SVM_AB: A vs B
 - SVM_AC: A vs C
 - SVM_BC: B vs C
- Each SVM casts a vote; the class with the most votes is selected.

Summary

- SVM key points:
 - Finds an optimal separating hyperplane.
 - Maximizes margin between classes.
 - Uses support vectors only.
 - Extends to nonlinear problems via kernel trick.

